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Assignment on Measurement and Instrumentation

$$X = 6 \quad Y = 4 \quad XY = 64 \quad X+Y = 10 \quad X \times Y = 24$$

Assignment 1

True value : 100.0 units.

Instrument reading : 106.4 units.

$$\begin{aligned} \text{Absolute error} &= 106.4 - 100.0 \\ &= 6.4 \end{aligned}$$

$$\begin{aligned} \text{Percentage error} &= \frac{106.4 - 100.0}{100.0} \times 100\% \\ &= 6.4\% \end{aligned}$$

Assignment 2

R 	R_s 	$P = I^2 R$
$E = 10V \pm 0.1V (1\%)$	$E_s = 1.2V \pm 0.064V$	$V = IR$
	$R_s = 0.0646 \Omega \pm 0.25\%$	$P = VI$

$$P = VI \quad \text{---} \quad \text{①}$$

$$I = \frac{V_s}{R_s} \quad \text{---} \quad \text{②}$$

$$\text{②} \rightarrow \text{①} \quad P = V \left(\frac{V_s}{R_s} \right) \quad \text{---} \quad \text{③}$$

$$P = 10V \left(\frac{1.2V}{0.0646 \Omega} \right)$$

$$P = 185.7585 W$$

$$\begin{aligned} \frac{W_R}{R} &= \left[\left(\frac{\partial E}{E} W_E \right)^2 + \left(\frac{\partial E_s}{E_s} W_{E_s} \right)^2 + \left(\frac{\partial R_s}{R_s} W_{R_s} \right)^2 \right]^{1/2} \\ &= \left[\left(\frac{(1)(0.1)}{10} \right)^2 + \left(\frac{(1)(0.064)}{1.2} \right)^2 + \left(\frac{(-1)(0.0646 \times 0.25\%)}{0.0646} \right)^2 \right]^{1/2} \end{aligned}$$

$$\frac{W_R}{R} = 0.0543$$

$$W_R = 0.0543 (185.7585)$$

$$W_R = 10.0904 W$$

$$\therefore P = 185.76 W \pm 10.09 W$$

Assignment 3

$$R_1 = (100 \pm 20) \Omega$$

$$R_2 = (50 \pm 30) \Omega$$

$$\begin{array}{c} R_1 \\ \text{---}\text{---}\text{---} \\ R_1 = 120 \Omega \pm 5\% \end{array}$$

$$\begin{array}{c} R_2 \\ \text{---}\text{---}\text{---} \\ R_2 = 100 \Omega \pm 5\% \end{array}$$

$$R_{\text{total}} = 120 \Omega + 100 \Omega$$

$$= 220 \Omega$$

$$W_R = \sqrt{(120 \times 5\%)^2 + (100 \times 5\%)^2}$$

$$= \sqrt{6^2 + 5^2}$$

$$W_R = 7.8102 \Omega$$

$$\therefore R_{\text{total}} = 220 \Omega \pm 7.81 \Omega$$

Assignment 4

$$L_1 = 10.0 + 0.6 = 10.6$$

$$L_2 = 10.2 + 0.4 = 10.6$$

$$L_3 = 9.8 + 0.6 = 10.4$$

$$L_4 = 10.1 + 0.4 = 10.5$$

$$\bar{x} = \frac{10.6 + 10.6 + 10.4 + 10.5}{4}$$

4

$$\bar{x} = 10.525$$

	x	$x - \bar{x}$	$(x - \bar{x})^2$
L_1	10.6	0.075	0.0056
L_2	10.6	0.075	0.0056
L_3	10.4	-0.125	0.0156
L_4	10.5	-0.025	0.00062
Σ			0.0275

$$\sigma = \sqrt{\frac{0.0275}{4-1}}$$

$$\sigma = 0.0957$$

Assignment 5

T = 20 + (25)

$$R_T = R_0 [1 + \alpha (T - 20^\circ\text{C})]$$

$$R_0 = 164 \Omega \pm 1\% \text{ , } 20^\circ\text{C}$$

$$W_{R_0} = 164 \Omega \times 1\% = 1.64 \Omega$$

$$\alpha = 0.004 \text{C}^{-1} \pm 1\%$$

$$W_\alpha = 0.004 \text{C}^{-1} \times 1\% = 0.00004 \text{C}^{-1}$$

$$T = 45^\circ\text{C} \pm 0.5^\circ\text{C}$$

$$W_T = 45^\circ\text{C} \times 0.5\% = 0.225^\circ\text{C}$$

$$R_T = 164 \Omega [1 + 0.004 \text{C}^{-1} (45^\circ\text{C} - 20^\circ\text{C})]$$

$$R_T = 180.4 \Omega$$

$$R_T = R_0 [1 + \alpha (T - 20^\circ\text{C})]$$

$$\frac{\partial R_T}{\partial R_0} = 1 + \alpha (T - 20^\circ\text{C})$$

$$\frac{\partial R_T}{\partial R_0} = 1 + 0.004 (45 - 20)$$

$$\frac{\partial R_T}{\partial R_0} = 1.1$$

$$R_T = R_0 + R_0 \alpha (T - 20^\circ\text{C})$$

$$\frac{\partial R_T}{\partial \alpha} = R_0 (T - 20^\circ\text{C})$$

$$\frac{\partial R_T}{\partial \alpha} = 164 (45 - 20)$$

$$\frac{\partial R_T}{\partial \alpha} = 4100$$

$$R_T = R_0 [1 + \alpha (T - 20^\circ\text{C})]$$

$$R_T = R_0 + R_0 \alpha T - R_0 \alpha (20^\circ\text{C})$$

$$\frac{\partial R_T}{\partial T} = R_0 \alpha$$

$$\frac{\partial R_T}{\partial T} = 164 (0.004)$$

$$\frac{\partial R_T}{\partial T} = 0.656$$

$$W_R = \left[\left(\frac{\partial R}{\partial R_0} (W_{R_0}) \right)^2 + \left(\frac{\partial R}{\partial \alpha} (W_\alpha) \right)^2 + \left(\frac{\partial R}{\partial T} (W_T) \right)^2 \right]^{1/2}$$

$$W_R = [(1.1 (1.64))^2 + (4100 (0.00004))^2 + (0.656 (0.225))^2]^{1/2}$$

$$W_R = 1.8174 \Omega$$

$$\therefore R_T = 180.4 \Omega \pm 1.82 \Omega$$

Assignment 6

$$R_{\text{series}} = \frac{V_{\text{max}}}{I_{\text{fs}}} - R_g$$

$$R_g = 100 \Omega$$

$$I_{\text{fs}} = 1 \text{ mA}$$

$$V_{\text{max}} = 84 \text{ V}$$

$$R_{\text{series}} = \left(\frac{84 \text{ V}}{1 \times 10^{-3} \text{ A}} \right) - 100 \Omega$$

$$R_{\text{series}} = 83900 \Omega$$

Assignment 7

$$N = 8 \text{ bits}$$

$$V = 16.4 \text{ V}$$

$$64 / 10 + 10 = 16.4$$

$$\text{Resolution} = \frac{16.4 \text{ V}}{2^8}$$

$$= 0.064 \text{ V-per step}$$

Assignment 8

$$\text{Time base} : 21 \text{ ms/div}$$

$$\text{One full cycle} : 12 \text{ horizontal divisions.}$$

$$\text{Period} : \frac{21 \text{ ms}}{1 \text{ div}} \times \frac{12 \text{ div}}{1 \text{ cycle}}$$

$$= 252 \text{ ms/cycle}$$

$$f = \frac{1}{\text{period}}$$

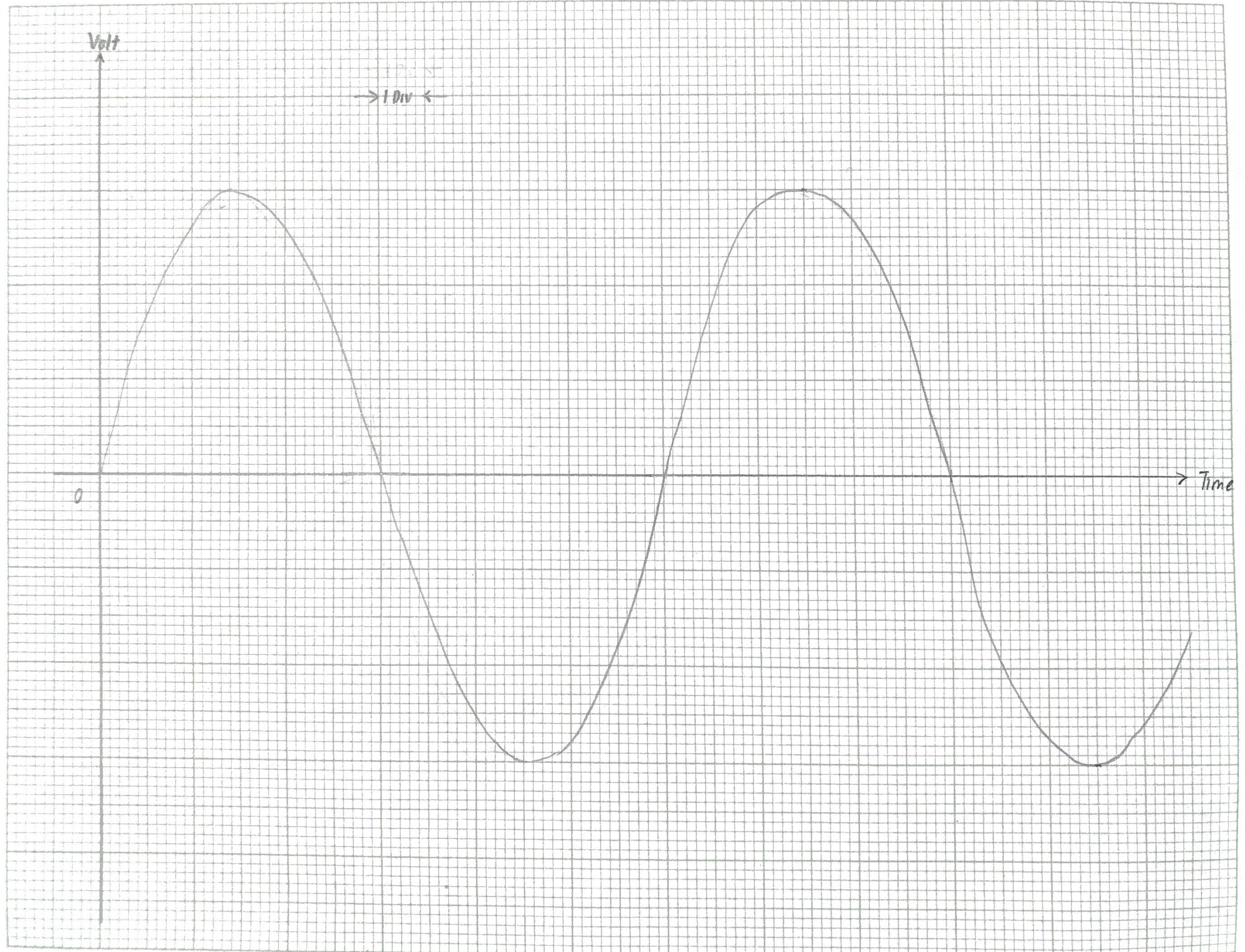
$$= \frac{1}{252 \times 10^{-3}}$$

$$= 3.968 \text{ s}^{-1}$$

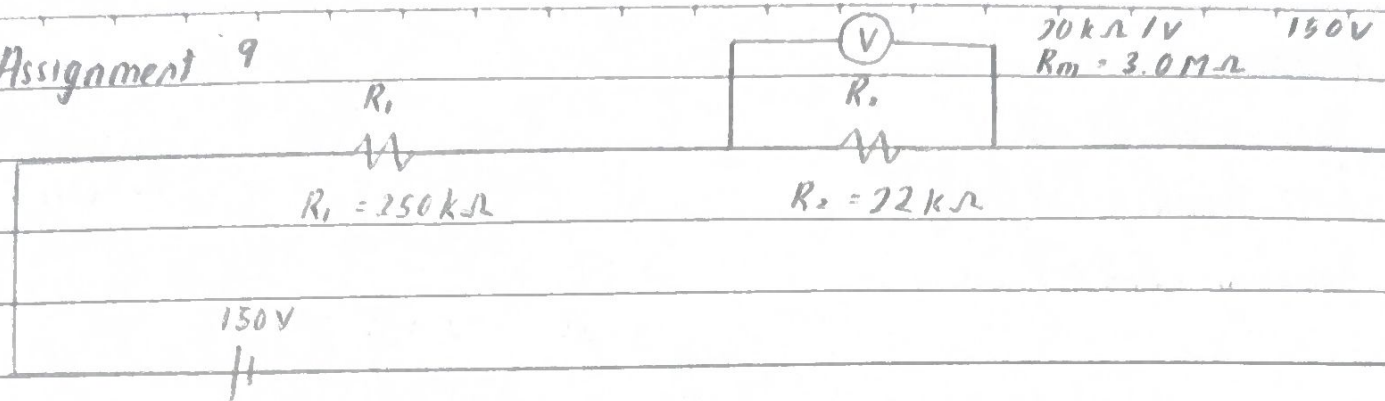
$$= 3.968 \text{ Hz}$$

Date:

No.



Assignment 9



$$R_{eq} = \frac{R_2 R_m}{R_2 + R_m}$$

$$R_{eq} = \frac{22 \times 10^3 \times 3 \times 10^6}{22 \times 10^3 + 3 \times 10^6}$$

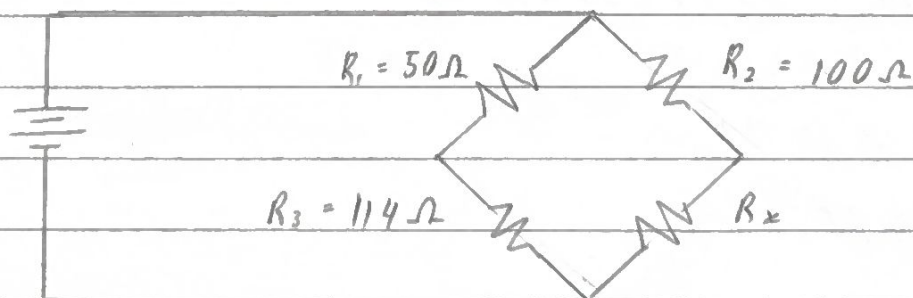
$$R_{eq} = 21839.84 \Omega$$

$$V_2 = V_s \left(\frac{R_{eq}}{R_1 + R_{eq}} \right)$$

$$V_2 = 150 \text{ V} \left(\frac{21839.84}{250 \times 10^3 + 21839.84} \right)$$

$$V_2 = 12.05 \text{ V}$$

Assignment 10



$$R_x = \frac{114 \Omega \times 100 \Omega}{50 \Omega}$$

$$R_x = 228 \Omega$$

Assignment 11

$$V_{\text{see}} = 18.68 \text{ mV}$$

$$\frac{64.11}{10} = 6.5$$

Cold junction: 0°C

$$V_{\text{meas}} = 6.5 \text{ mV}$$

$$\text{Sensitivity} = \frac{18.68 \text{ mV}}{500^\circ\text{C}}$$

$$500^\circ\text{C}$$

$$= 0.03736 \text{ mV}/^\circ\text{C}$$

$$T = \frac{6.5 \text{ mV}}{18.68 \text{ mV}} \times 500^\circ\text{C}$$

$$18.68 \text{ mV}$$

$$T = 173.98^\circ\text{C}$$

Assignment 12

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$A = 74 \times 10^{-4} \text{ m}^2$$

$$d = 1.0 \times 10^{-3} \text{ m}$$

$$\epsilon_r = 1.0$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$C = \frac{8.854 \times 10^{-12} (1.0) (74 \times 10^{-4})}{1.0 \times 10^{-3}}$$

$$1.0 \times 10^{-3}$$

$$C = 6.55196 \times 10^{-11} \text{ F}$$

$$C = 65.5196 \text{ pF}$$

Assignment 13

$$\text{Sensitivity} = 2.00 \text{ V/mm}$$

$$\text{Displacement} = 5.5 \text{ mm}$$

$$\begin{aligned} V_{out} &= 2.00 \text{ V/mm} \times 5.5 \text{ mm} \\ &= 11 \text{ V} \end{aligned}$$

Given the small displacement, this may not be the peak and can assume that it is still linear.